

EC 9540: HUMAN COMPUTER INTERACTION
MINI GROUP PROJECT
LAB 03



YamuLK

**A Smart Travel Planning Platform for Sri Lankan
Domestic Travelers**

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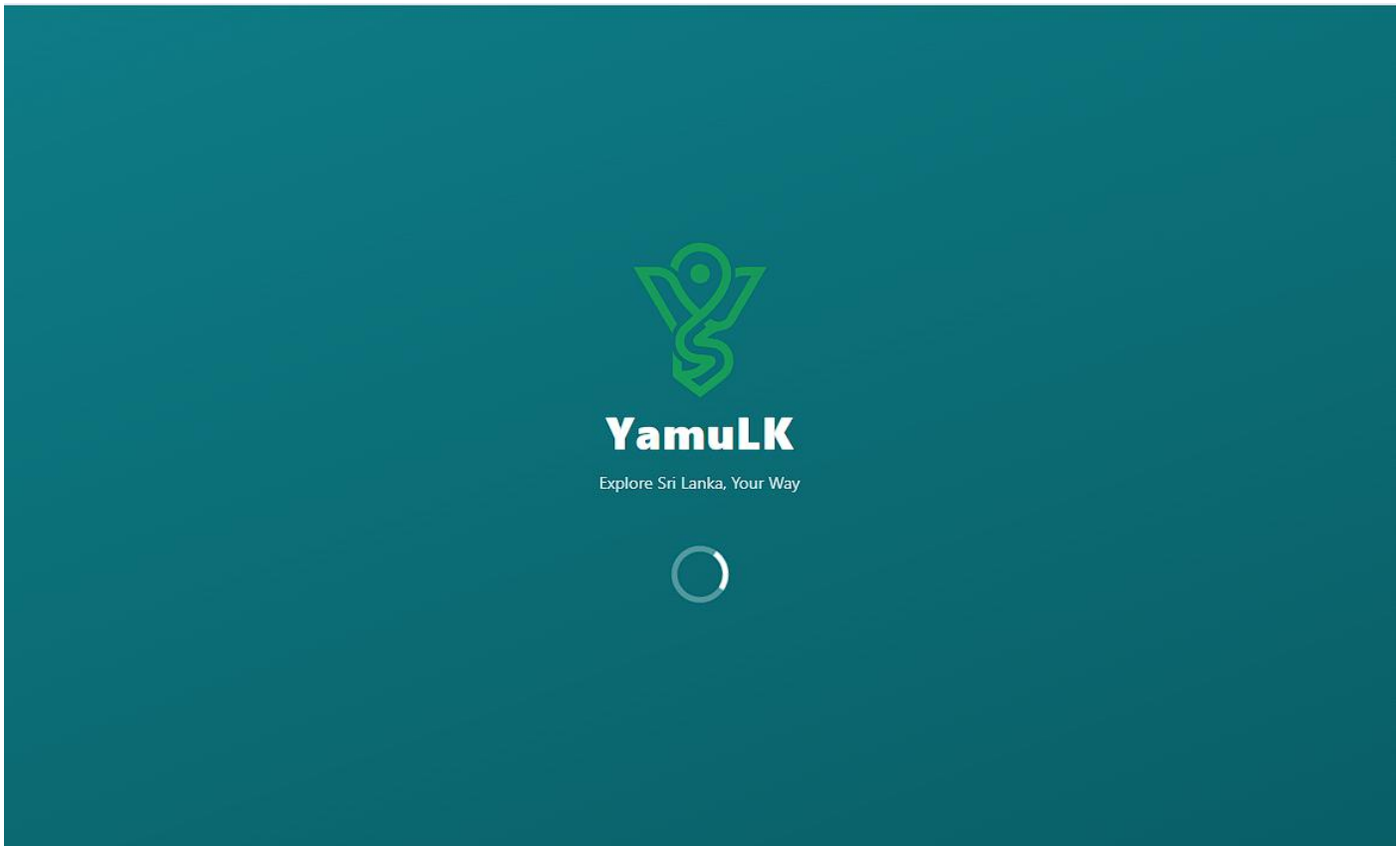
Introduction

Based on the user requirements identified in Lab 02, this lab focuses on designing the overall structure of the YamuLK mobile application. The aim is to create an interface that enables Sri Lankan domestic travelers to efficiently search destinations, estimate travel costs, generate travel itineraries, find accommodations, view weather updates, and plan routes using a single platform.

The system design includes the user interfaces, menu structure, graphical user interface (GUI) elements, navigation flow, back-end design, and the overall architecture of the application.

Interface Design

Splash Screen:



Login / Guest Screen:



Discover Sri Lanka's Hidden Gems

Plan trips, track budgets, and explore beaches, waterfalls, mountains and historical sites — all in one place.

Welcome Back

Sign in to continue planning your journey

Username

Password

Login

Register

OR

[Continue as Guest](#)

Home Screen:

**YamuLK**

HomeExploreSaved TripsTrip Planner



Categories

**Beaches**

**Waterfalls**

**Mountains**

**Historical Places**

Recommended for You

**Mirissa Beach**

Southern Province

★ 4.7 · Rs. 3,500/day

**Diyaluma Falls**

Uva Province

★ 4.8 · Rs. 2,000/day

**Ella Rock**

Uva Province

★ 4.9 · Rs. 1,500/day

[See all](#)

Popular Historical Sites



Sigiriya Rock

Central Province

★ 4.9 · Rs. 5,000/day



Anuradhapura

North Central Province

★ 4.6 · Rs. 2,500/day



Polonnaruwa

North Central Province

★ 4.7 · Rs. 2,200/day

Destination Details Screen:



YamuLK

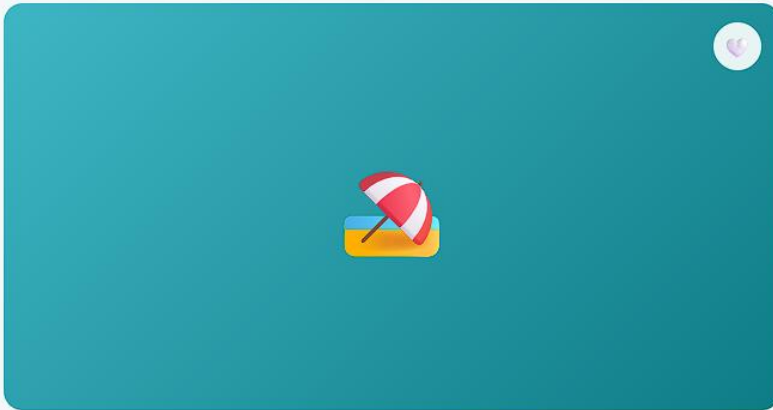
Home

Explore

Saved Trips

Trip Planner

KP



ESTIMATED COST

Rs. 3,500/day

WEATHER

☀️ 29°C

Nearby Hotels

🏠 Mirissa Bay Villa

🏠 Paradise Beach Club

Nearby Restaurants

🍽️ Zephyr Restaurant

🍽️ Dewmini Roti Shop

Nearby Attractions

🐳 Whale Watching Point

🌴 Coconut Tree Hill



Mirissa Beach

★ 4.7 (1.2k reviews) 📍 168 km from Colombo

A tranquil coastal escape famous for golden sands, whale watching tours, and stunning sunsets. One of the most visited beach towns in Southern Sri Lanka, ideal for weekend getaways and honeymooners alike.

Plan Trip

📌 Save

📢 Share

Trip Planner:

 **YamuLK**

HomeExploreSaved TripsTrip Planner

KP

Plan Your Trip

Minissa Beach · Southern Province

Travel Date

Number of People

Budget (Rs.)

Transport Method

 Bus

 Train

 Car

 Bike

Generate Plan

Budget Calculator:

 **YamuLK**

HomeExploreSaved TripsTrip Planner

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Budget Calculator

ESTIMATED TOTAL COST

Rs. 13,800

of Rs. 15,000 budget

✓ Within Budget

 Transport

4000

 Accommodation

6000

 Food

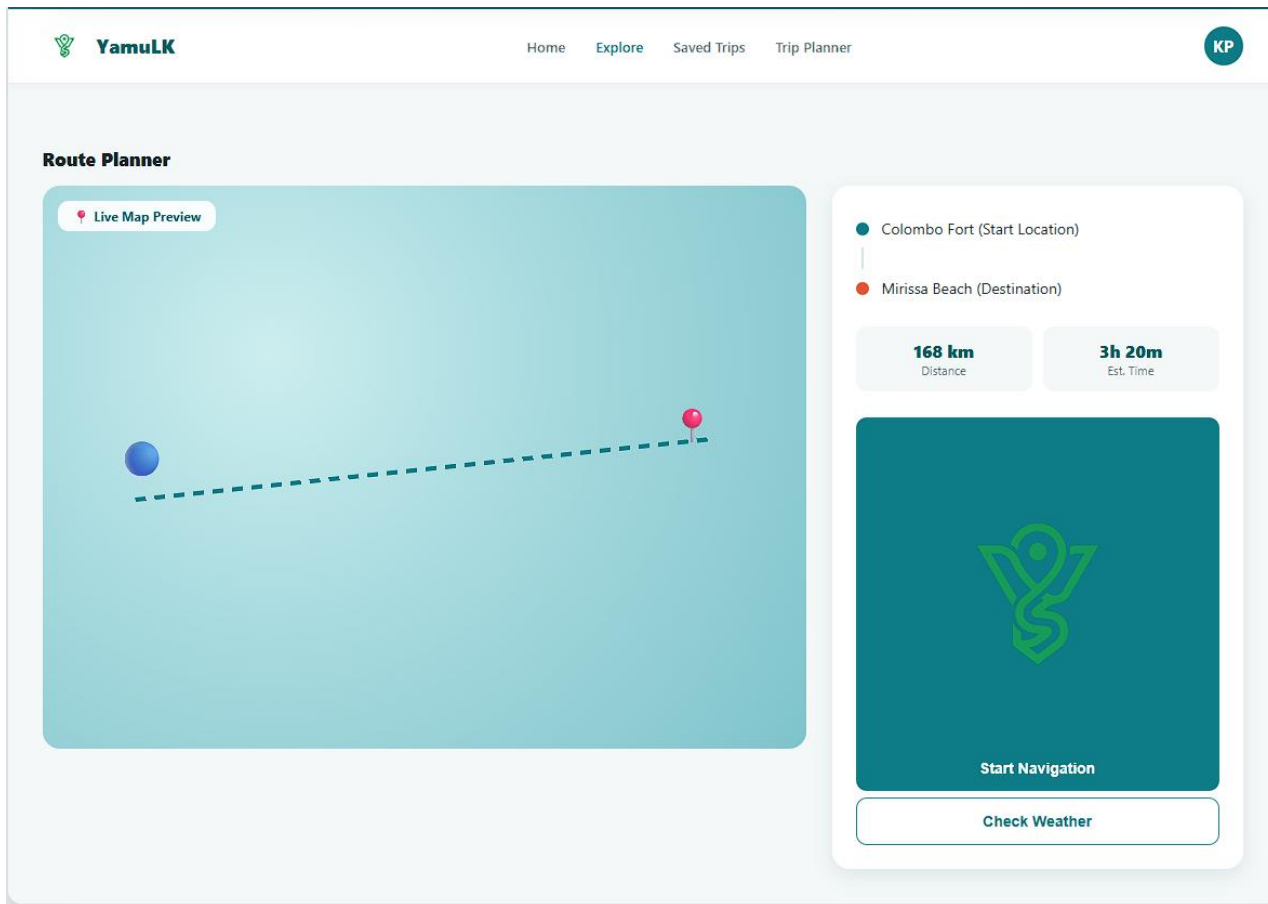
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 Other Expenses

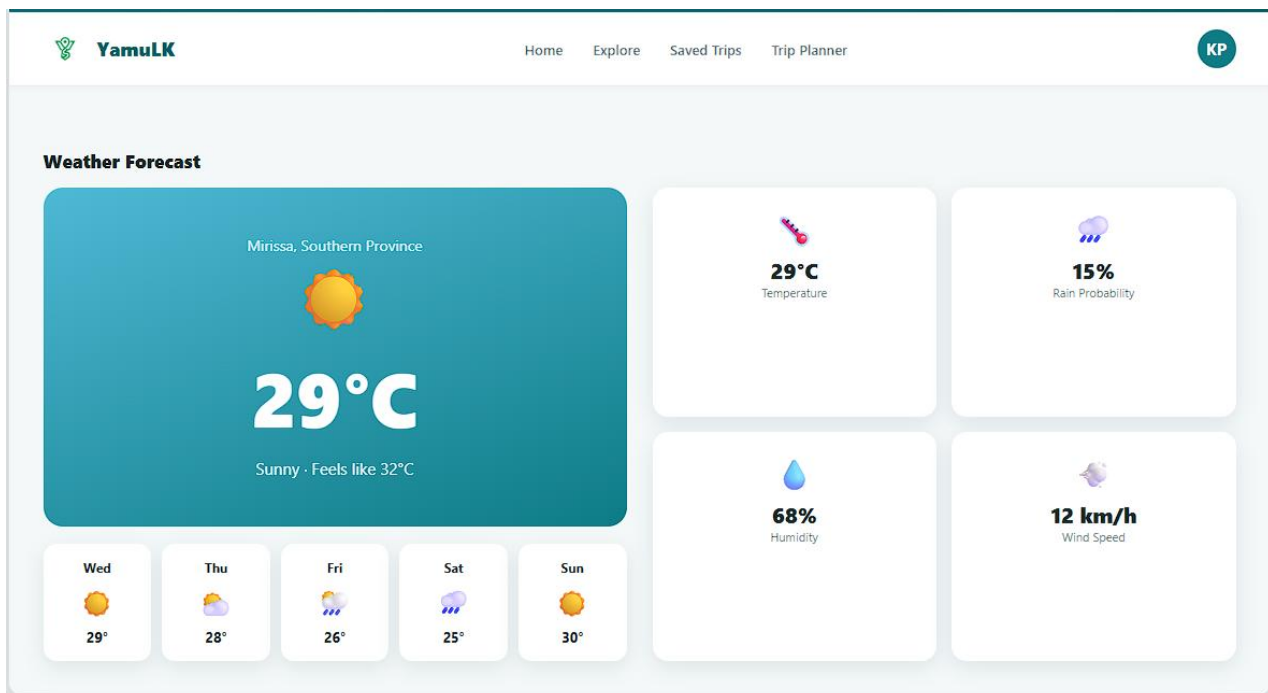
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View Route

Route Planner:



Weather:



Saved Trips:

 **YamuLK**

HomeExploreSaved TripsTrip Planner

KP

Saved Trips

Saved Itineraries

Favourite Destinations



Mirissa Weekend Trip
2 people · Rs. 13,800 · 2 days



Ella Hiking Adventure
4 people · Rs. 22,500 · 3 days



Sigiriya Rock Fortress
Favourite · Central Province



Diyaluma Falls
Favourite · Uva Province

Profile:

 **YamuLK**

HomeExploreSaved TripsTrip Planner

KP

My Profile



KP

Kavindu Premarathna
kavindu.premarathna@email.com

Edit Profile

 User Details

 Preferences

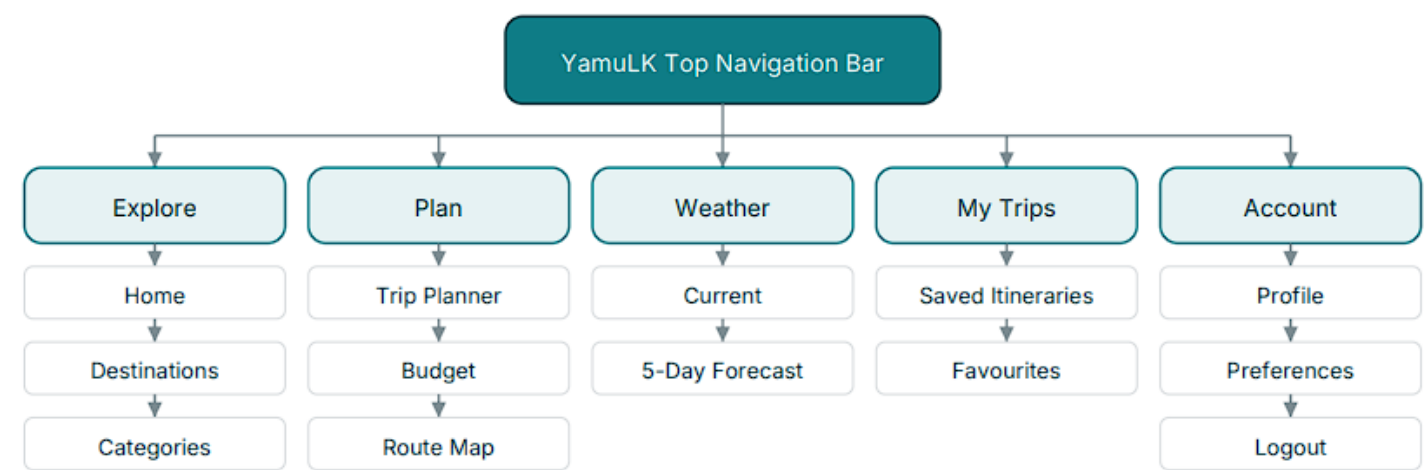
 Saved Trips

 Notifications

 Help & Support

 Logout

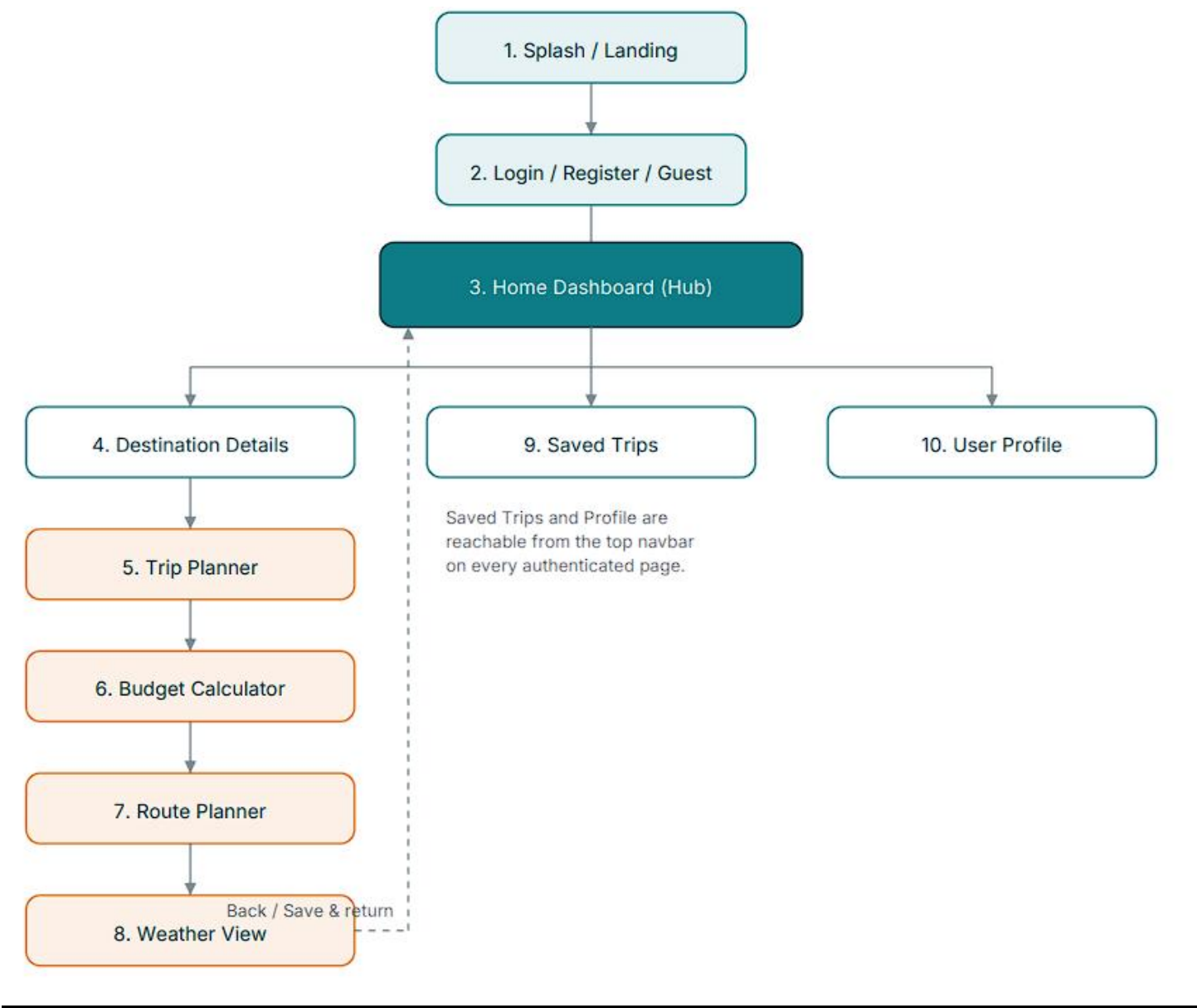
Menu Structure



GUI Elements

Screen	GUI Elements
Splash screen	Logo, Label, Loading Indicator
Login screen	Labels, Text Fields, Password Field, Buttons
Home screen	Search Bar, Cards, Images, Icons, Navigation Bar
Destination details	Images, Labels, Buttons, Rating Bar
Trip planner	Date Picker, Dropdown Menu, Text Fields, Buttons
Budget calculator	Number Fields, Labels, Calculate Button
Route planner	Map View, Buttons, Labels
Weather screen	Icons, Labels, Cards
Saved trips	List View, Cards, Buttons
Profile screen	Text Fields, Profile Image, Buttons

Navigation Flow



Back-end Plan

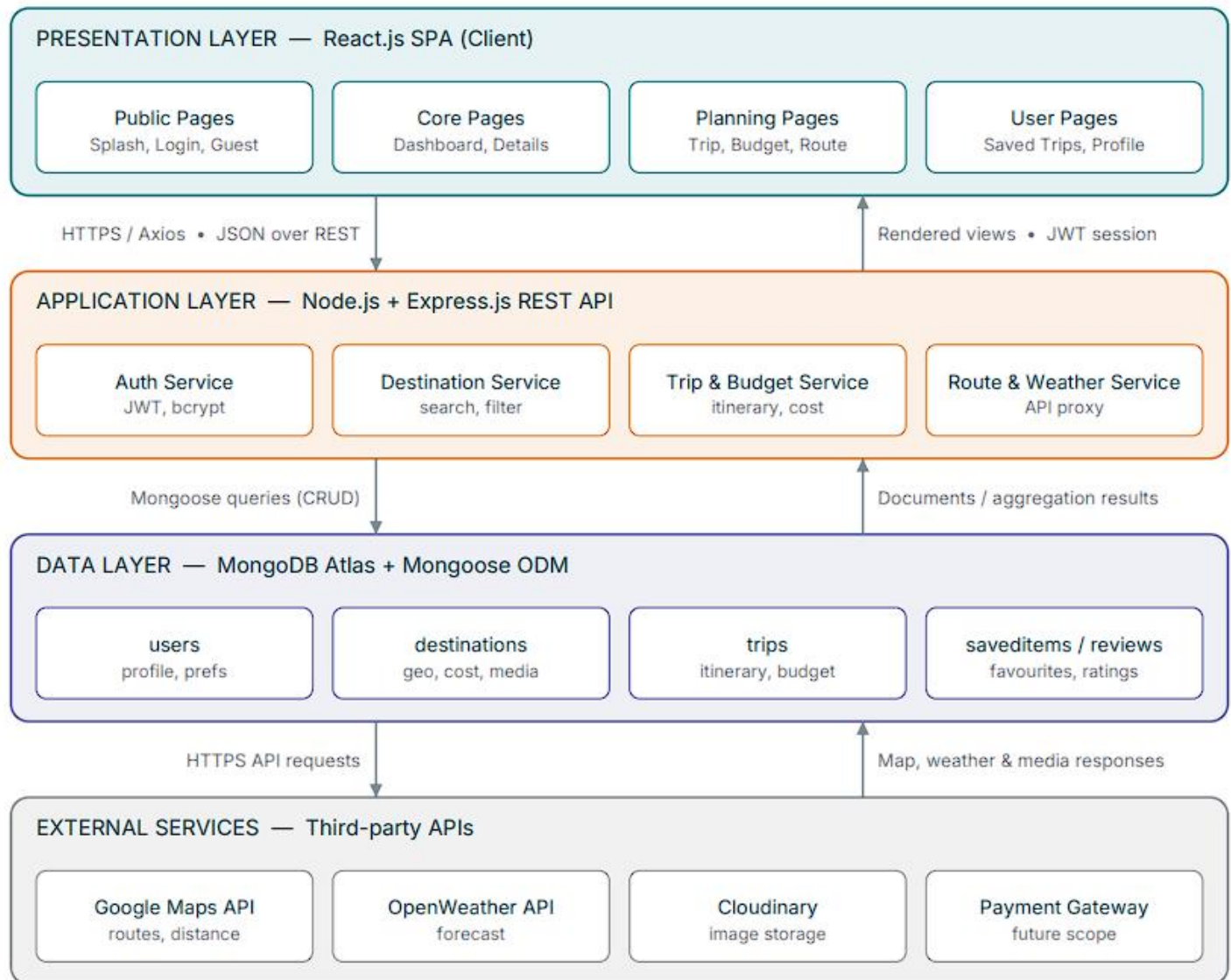
Required data and collection:

Collection	Key Field	Notes
users	_id, name, email (unique), passwordHash, preferences {categories, currency, homeProvince}, savedTrips [ObjectId], createdAt	Passwords hashed with bcrypt; email carries a unique index.
destinations	_id, name, category, province, description, images [url], location {lat, lng}, averageRating, estimatedCost, popularity	Text index on name and description for search; 2dsphere index on location
trips	_id, userId, destinationId, travelDate, people, transportMode, itinerary [{day, activities}], budgetId, status	One trip document per planned journey; references the user and the destination.
budgets	_id, tripId, transport, accommodation, food, other, totalCost, userBudget, status	Status is a derived value: Within Budget, Near Limit or Over Budget.
savedItems	_id, userId, itemType (destination or trip), itemId, savedAt	Compound unique index on userId and itemId prevents duplicates.
reviews	_id, userId, destinationId, rating, comment, createdAt	Aggregation pipeline recalculates the destination average rating.

Major Back-end Functions and API Endpoints:

Service module	Endpoint	Function	Description
Auth	POST /api/auth/register	registerUser()	Validates input, hashes the password, and creates the user document.
Auth	POST /api/auth/login	loginUser()	Verifies credentials and returns a signed JWT.
Auth	GET /api/auth/me	getCurrentUser()	Returns the profile of the token holder.
Destination	GET /api/destinations	listDestinations()	Supports search, category, province, and sort query parameters with pagination.
Destination	GET /api/destinations/:id	getDestinationById()	Returns full detail plus nearby hotels, restaurants, and attractions.
Trip	POST /api/trips	generateItinerary()	Builds a day-by-day plan from date, party size, budget, and transport mode.
Trip	GET /api/trips	getUserTrips()	Lists all itineraries belonging to the logged-in user.
Trip	PUT /api/trips/:id	updateTrip()	Edits an existing itinerary.
Trip	DELETE /api/trips/:id	deleteTrip()	Removes an itinerary and its budget document.
Budget	POST /api/budget/calculate	calculateBudget()	Sums the expense categories and returns the total.
Budget	GET /api/budget/:tripId	checkBudgetStatus()	Compares total cost against the user budget and returns the status label.
Route	GET /api/route	getRoute()	Calls the Google Maps Directions API and returns distance, duration, and polyline.
Weather	GET /api/weather/:destId	getWeather()	Returns cached or freshly fetched current conditions and a five-day forecast.
Saved	POST /api/saved	saveItem()	Adds a destination or trip to the user's saved list.
Saved	DELETE /api/saved/:id	removeSavedItem()	Removes an entry from the saved list.
Profile	PUT /api/users/profile	updateProfile()	Update's name, preferences, and password.
Review	POST /api/reviews	addReview()	Stores a review and recalculates the destination average rating.

Overall System Model



Conclusion

The planned system design provides a clear blueprint for implementing the YamuLK travel planning application. The interface design emphasizes simplicity and ease of use, while the navigation flow ensures users can efficiently access the application's core features. The proposed back-end structure supports essential functionalities such as destination search, itinerary generation, budget estimation, weather updates, accommodation recommendations, and travel plan management. This design will serve as the foundation for the front-end and back-end implementation phases in the upcoming labs.